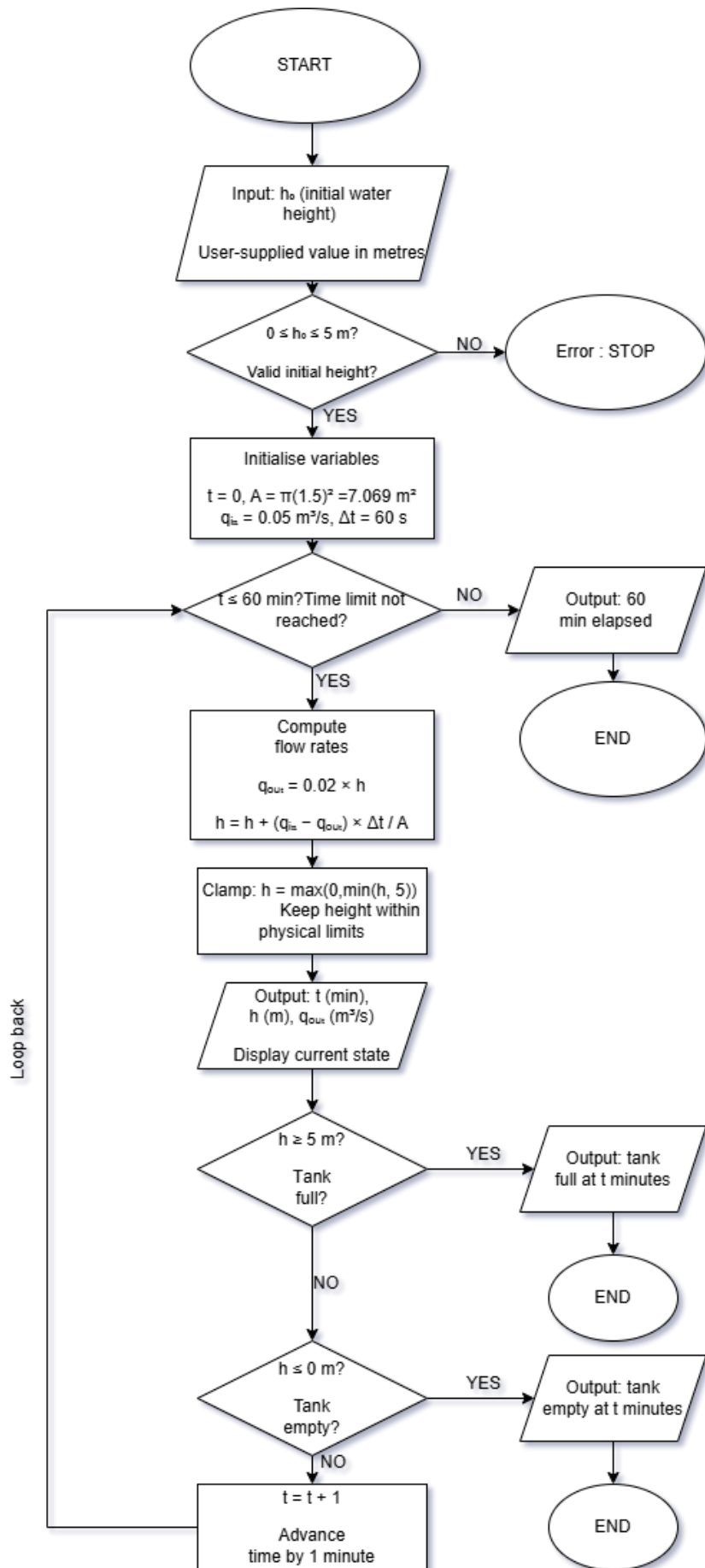


COMPUTER PROGRAMMING FOR PETROLEUM ENGINEERS

INDEX NUMBER: 4072924

PROJECT 1 -**Flowcharts & Pseudocode**

PART A



PART B – Pseudocode

Water Tank Simulation Algorithm

ALGORITHM SimulateWaterTank

INPUT h // User enters initial water height (m)

IF h < 0 **OR** h > 5 **THEN**

OUTPUT "Error: initial height must be between 0 and 5 m"

END IF

SET t = 0 // elapsed time in minutes

SET A = 3.142 * (1.5 * 1.5) // cross-sectional area = 7.069 m²

SET q_{in} = 0.05 // pump inflow rate (m³/s)

SET dt = 60 // time step in seconds

WHILE t <= 60

SET q_{out} = 0.02 * h // outlet flow depends on height

SET q_{net} = q_{in} - q_{out} // net flow rate (m³/s)

SET h = h + (q_{net} / A) * dt // update water height

IF h > 5 **THEN**

SET h = 5

ELSE IF h < 0 **THEN**

SET h = 0

END IF

OUTPUT t, h, q_{out} // display current state

IF h >= 5 **THEN**

OUTPUT "Tank full at t = ", t, " min"

END ALGORITHM

END IF

IF h <= 0 **THEN**

OUTPUT "Tank empty at t = ", t, " min"

END ALGORITHM

END IF

SET t = t + 1 // advance by 1 minute

END LOOP

OUTPUT "60-minute limit reached"

END ALGORITHM

PART C- PATTERN ANALYSIS

Algorithmic Patterns in the Solution

In the initialization part, variables are initialized one after another in a pre-determined order. All are assigned before anything else runs, e.g. t , A , q_{in} , and dt . And there is also sequence in the steps for computing inside the loop: first q_{out} is computed, then q_{net} , and then h is updated. Order is important because each step depends on the step before it.

There are three instances of choice. The first is the first input validation, which checks that h is between 0 and 5 before the simulation can proceed. The second and third are in the loop. They are responsible for determining if the tank is full or empty. If either of these is true, the procedure terminates.

Each iteration is the WHILE loop that runs through the simulation once a minute. It is a pre-test loop, which means that the condition $t \leq 60$ is checked before the loop body is executed. The loop continues until one of the three termination conditions is satisfied.

(A) Analogous Engineering Problem

The similar pattern is used in wellbore pressure simulation during drilling. The algorithm loops over time step updating annular pressure(iteration) after the driller sets initial conditions like mud weight and pump rate(sequence). At each step, it determines whether the pressure has dropped below pore pressure or exceeded the fracture gradient(selection), stopping the simulation if either limit is violated. The water tank problem follows the same structure: initialize, loop, compute, check and stop.

B) Abstraction Decisions

Since they have no bearing on the stepping algorithm's logic, a number of real-world factors were purposefully disregarded:

Variation in pump delivery: q_{in} was assumed to be constant at $0.05 \text{ m}^3/\text{s}$. The back pressure caused by the rising water level really affects the pump's output.

Fluid characteristics: volume and height are always directly related because water density and temperature were taken to be constant.

Valve behavior: the outlet was modeled as being precisely proportionate to h and constantly open. The opening and shutting of a genuine valve are delayed.

Measurement error: The simulation makes the assumption that h is precisely known at each time step. There is latency and noise in real sensors.

Fixed time step: a steady 60-second interval was employed. To prevent overshooting, the time step would decrease close to the boundary conditions in a true engineering simulation.

While these factors are critical in a real tank design, ignoring them does not change the sequence-selection-iteration structure of the algorithm, and this is what this simulation is meant to show.